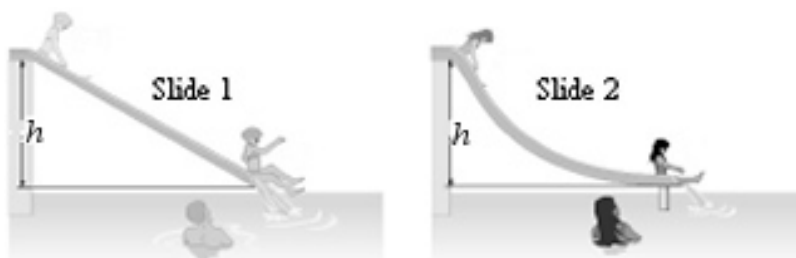


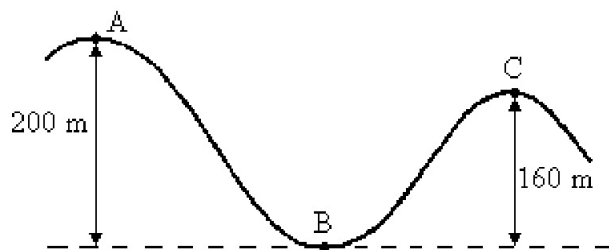
Name _____

- 1) A traveler pulls on a suitcase strap at an angle 36° above the horizontal. If 908 J of work are done by the strap while moving the suitcase a horizontal distance of 15 m, what is the tension in the strap?
- A) 61 N B) 85 N C) 75 N D) 92 N
- 2) Swimmers at a water park have a choice of two frictionless water slides as shown in the figure. Although both slides drop over the same height, h , slide 1 is straight while slide 2 is curved. How does the speed v_1 of a swimmer reaching the end of slide 1 compares with v_2 , the speed of a swimmer reaching the end of slide 2?



- A) $v_1 = v_2$
B) $v_1 < v_2$
C) $v_1 > v_2$
D) No simple relationship exists between v_1 and v_2 .
- 3) It requires 49 J of work to stretch an ideal very light spring from a length of 1.4 m to a length of 2.9 m. What is the value of the spring constant of this spring?
- A) 22 N/m B) 15 N/m C) 44 N/m D) 29 N/m
- 4) How much work is needed to change the speed of a 1600 kg vehicle from 15.0 m/s to 40.0 m/s?
- A) 40.0 kJ B) 10.0 kJ C) 0.960 MJ D) 1.10 MJ E) 20.0 kJ
- 5) A 1000 kg car experiences a net friction force of 9500 N while decelerating from 30.0 m/s to 23.4 m/s. How far does it travel while slowing down?
- A) 17.4 m B) 21.9 m C) 18.5 m D) 20.2 m

- 6) A bead is moving with a speed of 20 m/s at position A (height 200 m) on the track shown in the figure. This track is friction-free, and there is no appreciable air resistance. What is the speed of the bead at point C (height 160 m)?



- A) 20 m/s
 B) 34 m/s
 C) 0 m/s
 D) 69 m/s
 E) We cannot solve this problem without knowing the mass of the bead.
- 7) A 1500-kg car moving at 25 m/s hits an initially uncompressed horizontal ideal spring with spring constant of 2.0×10^6 N/m. What is the maximum distance the spring compresses?
 A) 0.17 m B) 0.34 m C) 0.68 m D) 0.51 m
- 8) A stone is held at a height h above the ground. A second stone with four times the mass of the first one is held at the same height. The gravitational potential energy of the second stone compared to that of the first stone is
 A) four times as much.
 B) twice as much.
 C) one-fourth as much.
 D) one-half as much.
 E) the same.
- 9) A rubber ball and a lump of clay have equal mass. They are thrown with equal speed against a wall. The ball bounces back with nearly the same speed with which it hit. The clay sticks to the wall. Which one of these objects experiences the greater momentum change?
 A) the clay
 B) the ball
 C) Both of them experience the same non-zero momentum change.
 D) Both of them experience zero momentum change.
- 10) During a collision with a wall, the velocity of a 0.200-kg ball changes from 20.0 m/s toward the wall to 12.0 m/s away from the wall. If the time the ball was in contact with the wall was 60.0 ms, what was the magnitude of the average force applied to the ball?
 A) 107 N B) 40.0 N C) 16.7 N D) 13.3 N E) 26.7 N
- 11) A 480-kg car moving at 14.4 m/s hits from behind a 570-kg car moving at 13.3 m/s in the same direction. If the new speed of the heavier car is 14.0 m/s, what is the speed of the lighter car after the collision, assuming that any unbalanced forces on the system are negligibly small?
 A) 19.9 m/s B) 13.6 m/s C) 10.5 m/s D) 5.24 m/s

- 12) In a perfectly ELASTIC collision between two perfectly rigid objects
- A) the kinetic energy of the system is conserved, but the momentum of the system is not conserved.
 - B) the momentum of the system is conserved but the kinetic energy of the system is not conserved.
 - C) both the momentum and the kinetic energy of the system are conserved.
 - D) the momentum of each object is conserved.
 - E) the kinetic energy of each object is conserved.
- 13) A 4.50-kg wheel that is 34.5 cm in diameter rotates through an angle of 13.8 rad as it slows down uniformly from 22.0 rad/s to 13.5 rad/s. What is the magnitude of the angular acceleration of the wheel?
- A) 111 rad/s²
 - B) 0.616 rad/s²
 - C) 5.45 rad/s²
 - D) 22.5 rad/s²
 - E) 10.9 rad/s²
- 14) While spinning down from 500.0 rpm to rest, a solid uniform flywheel (cylinder) does 5.1 kJ of work. If the radius of the disk is 1.2 m, what is its mass?
- A) 5.2 kg
 - B) 6.8 kg
 - C) 4.4 kg
 - D) 6.0 kg
- 15) When a rigid body rotates about a fixed axis, all the points in the body have the same
- A) tangential speed.
 - B) tangential acceleration.
 - C) centripetal acceleration.
 - D) linear displacement.
 - E) angular acceleration.